

Unit 13 : Internet and WWW

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Objectives

- understand IP security (IPsec)
- understanding Email Security
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- difference between Paid VPN and Free VPN
- understand digital signature.
- understand digital certificate.

Introduction

13.1 IP security (IPSec)

IP security (IPSec) is an Internet Engineering Task Force (IETF) standard suite of protocols that offer data authentication, integrity, and secrecy between two communication points over an IP network. It also specifies how packets are encrypted, decrypted, and authenticated. It specifies the protocols for safe key exchange and key management.

Uses of IP Security:

The following are some of the things IPsec may be used for.

- To encrypt data at the application layer.
- To ensure that routers delivering routing data across the public internet are secure.
- To give authentication without encryption, such as confirming that data comes from a recognized sender.

- To safeguard network data by establishing circuits utilizing IPsec tunnelling, in which all data transported between the two endpoints is encrypted, similar to a VPN connection.

Components of IP Security

It consists of the following elements:

1. Encapsulating Security Payload (ESP) -

Data integrity, encryption, authentication, and anti-replay are all included. It also supports payload authentication.

2. Authentication Header (AH) -

It also has data integrity, authentication, and anti-replay capabilities, but no encryption. Anti-replay protection guards against unwanted packet transfer. It does not ensure the privacy of data.

IP HDR	AH	TCP	DATA
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3. Internet Key Exchange -

It's a network security protocol that allows two devices to dynamically exchange encryption keys and discover a means to communicate across a Security Association (SA). To facilitate secure communication, the Security Association (SA) creates common security properties between two network entities. The Internet Security Association (ISA) and the Key Management Protocol (ISAKMP) offer a framework for authentication and key exchange. ISAKMP describes how Security Associations (SAs) are created and how direct connections between two IPsec-enabled hosts are established.

IKE (Internet Key Exchange) protects the content of messages and serves as an open framework for implementing common algorithms like SHA and MD5. Each packet is assigned a unique identification by the algorithm's IP sec users. The device may then determine if a packet is valid or not using this identification. Unauthorized packets are deleted and not delivered to the intended recipient.

IP HDR	TCP	DATA
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Original Packet

IP HDR	ESP HDR	TCP	Data	ESP Trailer	ESP Authentication
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←----- Encryption ---->

←----- Authentication ----->

Working of IP Security

- The host determines whether or not the packet should be sent via IPsec. These packets activate the security policy on their own. When the system transmitting the packet uses adequate encryption, this is accomplished. The host additionally verifies whether or not the incoming packets are correctly encrypted.
- Then there was IKE. Phase 1 begins with the two hosts (using IPsec) authenticating themselves to begin a secure communication. It has two different modes.

- The Main mode, which provides additional security, and the Aggressive option, which allows the host to quickly construct an IPsec circuit.
- Now, IKE Phase 2 takes place via a secure channel, with the two hosts negotiating the sort of cryptographic algorithms to employ and agreeing on secret keying material to use with those methods.
- The data is then sent via the newly established IPsec encrypted tunnel. The hosts use IPsec SAs to encrypt and decode these packets.
- When the communication between the hosts is finished or the session expires, the IPsec tunnel is ended by both hosts discarding the keys.

13.2 Email-Security

Email hacking may be accomplished in a variety of ways. Whether it's a spam virus or phishing. What exactly is spam? Sending unwanted mass emails to those who have not requested for them is known as email spamming.

Spam

Spam is junk mail delivered by commercial businesses as an advertising for their products and services. For such mails, we don't have an address. As a result, they are infected with the spam virus. Some emails may contain files containing harmful script, which, when performed on your computer, will mostly damage your vital data.

Phishing email

Phishing is when someone sends an email to a user that claims to be authentic but isn't. Its main goal is to steal sensitive data such as usernames, passwords, and credit card numbers.

Search emails contain a link to a malware-infected website that directs the user to input information on a phoney website that looks and feels real. So, kids, keep in mind one very crucial point:

Spams can cause a slew of issues:

- It floods your email account with unwanted emails, which may result in the loss of information emails; it consumes bandwidth, slowing the speed at which mails are delivered; some unsolicited emails may include viruses, which may harm your computer; it costs time and energy in reviewing and deleting junk emails or spam; it consumes bandwidth, slowing the speed at which mails are delivered; and it spends time and energy in reviewing and deleting junk emails or spam; and it consumes bandwidth, slowing the speed at which mail.

Blocking Spams

Use a different email address for sending messages to a newsgroup or mailing list than the one you use for personal correspondence. Your email address should not be posted on the internet since it might be easily spammed. Avoid responding to emails from unknown senders, and never buy anything in response to spam that promises a product, since these are all variables that might lead to identity theft.

How to clear up and archive your emails

You may go to the File tab and find the cleaning tools there, and if you click on the drop-down arrow, you'll see the choice of archives. Choose the option to archive this folder and all subfolders. Then, select the folder you wish to archive by clicking on it as seen in Figure 1. Choose an older date from the archived items. a list Now, select Browse to name and save a new dot PST file, and then click OK. As a result, you will be able to clean up and archive in this manner.

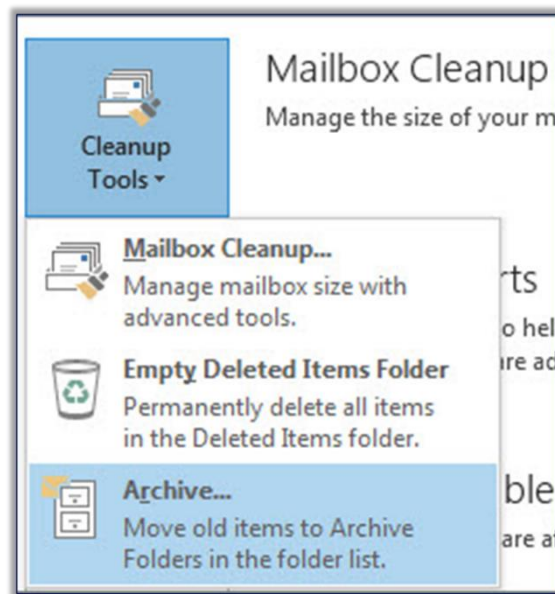


Figure 1 Email Cleanup

13.3 Virtual Private Network (VPN)

A virtual private network is the full version of the term VPN. What is a home network? You have an internet connection that you will use either to connect to the internet or to connect to the internet using a modem or a router. Let's look at how we are actually connecting to the internet.

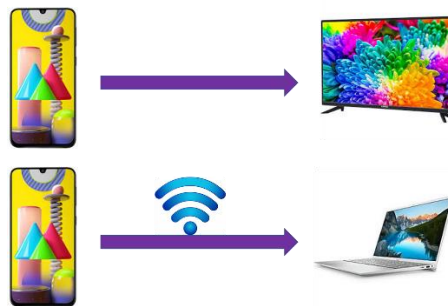


Figure 2 Internal Network

Assume we have a laptop, a television, and a cell phone. These three gadgets are present. Assume you want to send anything to your television, or you're working on your laptop. You're communicating by using a wireless connection that's connected to your phone and laptop as shown in Figure 2.

In a nutshell, these two types of networks are referred to as internal networks. This is your internal network if you're working on a laptop with a phone. Assume that any video you're seeing, such as the one you're seeing right now, is stored in a data centre. And that data centre may be in the United States, Canada, France, Germany, or somewhere else, so you could watch this movie from there. So any location implies you may access that movie; for example, if you are in India and want to view a film that is in Canada, Germany, or another country, you may do so. Let's look at Google as an example. Consider how they really request going to Google.com. Now you'll submit a request via a router, modem, or hub. Everyone has been given an address and will be working from that place.

If you wanted to open a google.com, you would send a request to the server, which would then be routed through the router. So the router has an address, and you can connect to the server, and the server responds to you, and you can sink the google.com domain.

What are the benefits of this sort of network in which IP addresses are routed to the server?

You will be able to connect to the Internet using this address.

As a result, the downsides include that your ISP service provider can track which websites you visit and what you track. Because your IP address is known by internet service providers. It's possible that your government is keeping an eye on you as well. When you're receiving anything from the server or uploading anything to the server, your data might be stolen. As a result, it's conceivable that your information will be stolen. We're relieved that we were able to connect to the open Wi-Fi networks. Your data may be encrypted at times. Hackers can access your data at any moment. A lot of stuff on the internet is confined to a specific region or nation. Assume you're sitting in India and you notice a video that's available in the United States or Canada, and you want to watch it. And such videos are banned on the internet, so you won't be able to see them if you're in a different nation.

In China, for example, you cannot access the Facebook website. You won't be able to register a Facebook account since China has already blocked it as seen in Figure 3. What is the greatest solution if you wish to utilise the Facebook site? As a result, we have a solution in the form of a VPN.



Figure 3 Banned in China

You may access any other site or programme that is on separate places on your machine by utilising a virtual private network. This is where you'll find the VPN box, and anything you're sending will be secured as well. Neither your internet service provider nor the government have any idea what you're talking about. If you make a request, the server assumes that you are both from the same country. It signifies you are receiving the response in accordance with the request. If someone tried to sniff your packet, it would be impossible for them to read it since it is encrypted. You can utilise content from another region as well as any website that is written in your area. You may also view using a virtual private network. It means you'll be able to access and utilise all information safely.

Definition of VPN

A virtual private network (VPN) is a sort of private network that communicates via public telecommunications, such as the Internet, rather than leased lines. It grew in popularity as more people worked in remote places and needed to learn how VPNs function.

Let's understand VPN by an example:

Consider a scenario in which a bank's headquarters are in Washington, D.C. This workplace has a local network with about 100 machines on it. Assume the bank has additional branches in Mumbai, India, and Tokyo, Japan. The old technique of creating a secure connection between the head office and the branch was to use a leased line between the branches and the head office, which was both expensive and inconvenient. We were able to effectively solve this problem thanks to VPN.

Some fantastic attributes are highlighted here:

- VPN also guarantees security by establishing a secure connection between the client and the VPN server.
- Many banned websites may be accessed with a VPN.

- VPN allows you to browse anonymously by masking your IP address.
- Also, the best Search engine optimization (SEO) is done by evaluating data from VPN providers that give country-by-country statistics on how people are using a given product. Many online marketing managers utilise this strategy of SEO to develop fresh ideas.

VPN and its legality

In most nations, using a VPN is legal. The legality of utilising a VPN service is determined by the country in question, as well as its geopolitical connections with other countries. A trustworthy and secure VPN is always lawful if you're not planning on using it for any unlawful purposes, such as online fraud, cyber theft, or downloading copyrighted information in some countries.

According to Bloomberg, China has determined to shut all VPNs (virtual private networks) starting next year. VPNs are used by many Chinese Internet users to access websites that are prohibited by China's so-called "great firewall." This is done to prevent information from being leaked to competing governments and to improve information security.

13.4 Difference between Free VPN and Paid VPN

The difference between Free VPN and Paid VPN is shown in Table 1.

Table 1 Difference between Paid VPN and Free VPN

	Free VPN	Paid VPN
Cost	Free	From USD 7 per month
Number of VPN Servers	Generally, up to 5-7	Generally, more than 40
Advertising	Yes	No
Connection speed	Low	High
Monthly Traffic Volume	Limited	Not Limited
Technical Support	No support or it is too low	Quick support, 24/7
Security guarantees	No	Yes
Personal servers	Never	Some developers provide
No logs policy	No guarantees	Yes

13.5 Digital Signature

Encryption: Process of transforming electronic data into cypher text, which is incomprehensible to everyone but authorised individuals. This ensures the safety of the data.

Decryption: The procedure for converting code into data.

- The message is encrypted at the sender's end using various encryption techniques, and then decoded at the receiver's end using decryption methods.
- Encryption and decryption techniques are employed to ensure data security when certain messages, such as usernames and passwords, must be kept secure.

Types of Encryption

1. **Symmetric Encryption:** A key is used to encrypt data, and the same key is used to decode it.

2. **Asymmetric Encryption:** Public key cryptography is another name for asymmetric cryptography. It encrypts and decrypts data using public and private keys. The public key is the one of the pair of keys that may be shared with anybody. The private key is the other key in the pair that is kept secret and only known by the owner. A message can be encrypted using one of the keys; The decryption key is the polar opposite of the one used to encrypt the communication.

Public Key

Everyone is aware of the key. The ex-public key of A is 7, and this fact is well known.

Private Key

Only the person whose private key it is is aware of it.

Authentication

Any procedure by which a system confirms the identity of a person who seeks to access it is known as authentication.

Non-repudiation

Non-repudiation refers to the process of ensuring that a transmitted communication was transmitted and received by the parties claiming to have transmitted and received it. Non-repudiation is a method of ensuring that the sender of a communication cannot later dispute sending it, and that the recipient cannot later dispute receiving it.

Integrity

to be sure the message wasn't tampered with during transmission

Message Digest

The representation of text as a single string of numbers, achieved with the use of a formula known as a one-way hash function. A digital signature is an electronic mechanism of authentication that is created by encrypting a message digest with a private key.

13.6 Digital Signature

A digital signature is a mathematical approach for verifying the integrity and validity of a communication, software, or digital document.

Key Generation Algorithms

Digital signatures are digital signatures that guarantee that a communication was transmitted by a certain sender. Authenticity and integrity should be ensured while conducting digital transactions; otherwise, data might be tampered with, or someone might pretend to be the sender and expect a response.

Signing Algorithms

Signing algorithms, such as email programmes, establish a one-way hash of the electronic data to be signed to establish a digital signature. The hash value is subsequently encrypted using the private key by the signing algorithm (signature key). The digital signature consists of this encrypted hash, as well as other information such as the hashing algorithm.

The data is appended with the digital signature and delivered to the verifier. Because a hash function turns any arbitrary input into a much shorter fixed length result, it's better to encrypt the hash rather than the full message or document. This saves time since a shorter hash value must now be signed instead of a big message, and hashing is significantly quicker than signing.

Signature Verification Algorithms

Along with the data, the Verifier obtains a Digital Signature. It then processes the digital signature and the public key (verification key) using the Verification algorithm and generates a value. It also creates a hash value by applying the same hash algorithm to the incoming data. The hash value and the verification algorithm result are then compared. If they're both equal, the digital signature is genuine; otherwise, it's not.

The steps for establishing a digital signature are as follows:

- Message digest is computed by applying hash function on the message and then message digest is encrypted using private key of sender to form the digital signature. (digital signature = encryption (private key of sender, message digest) and message digest = message digest algorithm(message)).
- The message is then sent with a digital signature. (A message with a digital signature is sent).
- The digital signature is decrypted by the receiver using the sender's public key. (As only the sender possesses his private key, only the sender may encrypt using his private key, which can then be decrypted by the sender's public key, ensuring authenticity.)
- The message digest has now been delivered to the recipient.
- The message digest can be computed by the receiver from the message (actual message is sent with the digital signature).
- For integrity, the message digest calculated by the receiver and the message digest (obtained by decrypting a digital signature) must be the same.
- The message digest is calculated using a one-way hash function, which is a hash function in which computing the hash value of a message is simple but computing the message from the hash value is complex.

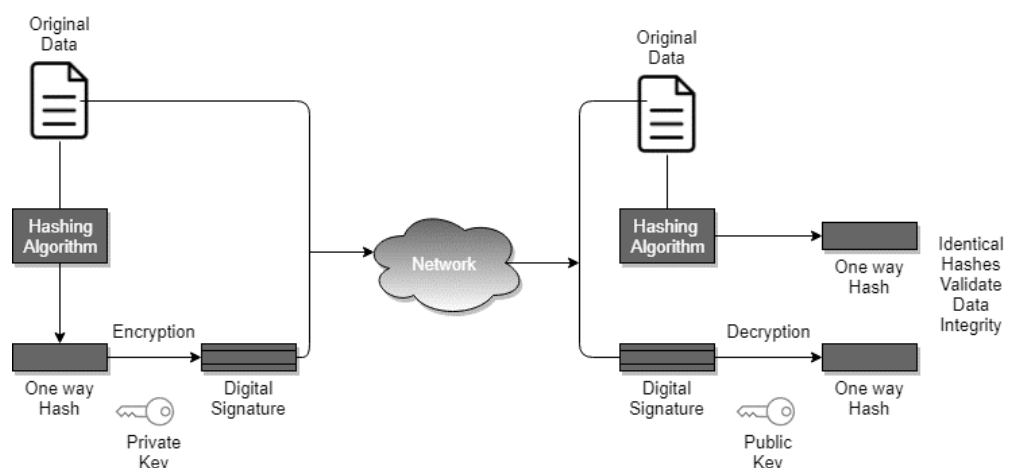


Figure 4 Digital Certificate

A trustworthy third party issues a digital certificate that validates the sender's identity to the receiver and the recipient's identity to the sender.

A certificate issued by a Certificate Authority (CA) to validate the identity of the certificate holder is known as a digital certificate. The CA creates an encrypted digital certificate that includes the applicant's public key and other identifying information. A digital certificate is used to associate a public key with a specific person or business.

Digital Certificate contains

1. Name of certificate holder.
2. Serial number which is used to uniquely identify a certificate, the individual or the entity identified by the certificate
3. Expiration dates.
4. Copy of certificate holder's public key.(used for decrypting messages and digital signatures)
5. Digital Signature of the certificate issuing authority.

Digital certificate is also sent with the digital signature and the message.

13.7 Digital certificate vs Digital signature

The difference between digital signature and digital certificate is shown in Table 2.

Table 2 Digital signature vs Digital certificate

Digital signature	Digital certificate
A digital signature is a string of decimals that is affixed to a file to assist with identifying the signer and ensuring its integrity	A digital certificate is itself a file that is used to assert identity and to facilitate encrypted connections.
Digital signature must ensure that the data or information remains secure from the moment it is sent	Digital certificate is that the certificate binds the digital signature to the object
Digital signatures are used to validate the sent data.	Digital certificates are used to validate the identity of the sender,
It follows Digital Signature Standard (DSS).	It follows X.509 Standard Format

Summary

- Internet Protocol Security (IPsec) is a secure network protocol suite that authenticates and encrypts data packets to offer safe encrypted communication between two computers over an IP network. It's a protocol that's utilised in virtual private networks.
- Any technique that secures email content and accounts from unwanted access is referred to as email security. Email security procedures are in place at email service providers to protect customer accounts and information from hackers.
- A virtual private network (VPN) creates a secure connection between you and the internet. All of your data traffic is routed over an encrypted virtual tunnel via the VPN. When you access the internet, this masks your IP address, making its location opaque to everyone. External assaults are also protected by a VPN connection.
- A digital signature is a mathematical approach for verifying the integrity and validity of a communication, software, or digital document. It's the digital counterpart of a handwritten signature or a stamped seal, but it has a lot more security built in.
- A public key certificate is an electronic document used to confirm the ownership of a public key in cryptography. It is also known as a digital certificate or an identity certificate.
- Encryption is the process of encoding data in cryptography. This technique transforms plaintext into ciphertext, which is an alternate representation of the information.
- Decryption is the process of restoring encrypted data to its original state. In most cases, it's a reversal of the encryption process. Because decryption requires a secret key or password, it decodes the encrypted information so that only an authorised user may decrypt the data.
- Spamming is the practise of sending many unsolicited messages to a large number of people for the purpose of commercial advertising, non-commercial preaching, or any other unlawful purpose, or just sending the same message to the same user over and over again.

Keywords

IP security: In IPv4 and IPv6 network packets, the IP security architecture (IPsec) offers cryptographic protection for IP datagrams.

internet key exchange: The mechanism used to establish up a security association in the IPsec protocol suite is known as Internet Key Exchange (IKE).

Email security: Any technique that secures email content and accounts from unwanted access is referred to as email security. Email security procedures are in place at email service providers to protect customer accounts and information from hackers.

Phishing: Phishing is a type of cybercrime in which a person acting as a genuine organisation contacts a target or targets by email, phone, or text message to persuade them to provide sensitive data such as personally identifying information, banking and credit card information, and passwords.

VPN: The term "virtual private network" refers to the ability to create a secure network connection while using public networks.

Digital signature: A digital signature is exactly what it sounds like: an electronic version of the traditional paper and pen signature.

Encryption: Encryption is the process of converting data into a secret code that hides the real meaning of the data.

Decryption: Decryption is the process of turning encoded/encrypted data into a form that a person or a machine can read and understand.

Public key: A public key is a huge numerical number used to encrypt data in cryptography. A software programme can produce the key, but it's more common for it to be issued by a trusted, recognised authority and made public through a publicly accessible repository or directory.

Private key: The owners keep their private keys hidden. Public keys are disseminated and used to validate credentials and authenticate nodes.

Authentication: The process of ascertaining if someone or something is who or what it claims to be is known as authentication.

Review Questions

1. Difference between Free VPN and paid VPN.
2. What is IP Security?
3. What is E-mail security?
4. Explain components of IP Security.
5. Write down steps for establishing a digital signature.
6. Write down difference between digital signature and digital certificate.
7. Explain types of signature.
8. What are the benefits of this sort of network in which IP addresses are routed to the server?
9. Explain working of IP Security
10. Difference between private key and public key.

Self- Assessment

1. IPSec is designed to provide security at the _____
 - a) Transport layer
 - b) Network layer
 - c) Application layer
 - d) Session layer
2. In tunnel mode, IPSec protects the _____
 - a) Entire IP packet

- b) IP header
 - c) IP payload
 - d) IP trailer
3. Which component is included in IP security?
- a) Authentication Header (AH)
 - b) Encapsulating Security Payload (ESP)
 - c) Internet key Exchange (IKE)
 - d) All of the mentioned
4. Which of them is not a major way of stealing email information?
- a) Stealing cookies
 - b) Reverse Engineering
 - c) Password Phishing
 - d) Social Engineering
5. The process of transforming plain text into unreadable text.
- a) Decryption
 - b) Encryption
 - c) Network Security
 - d) Information Hiding
6. A process of making the encrypted text readable again.
- a) Decryption
 - b) Encryption
 - c) Network Security
 - d) Information Hiding
7. A digital signature is a mathematical technique which validates?
- a) authenticity
 - b) integrity
 - c) Non-repudiation
 - d) All of the above
8. Which algorithm provides the private key and its corresponding public key?
- a) Key generation algorithm
 - b) Signature verifying algorithm
 - c) Signing algorithm
 - d) None of the above
9. The field that covers a variety of computer networks, both public and private, that are used in everyday jobs.
- a) Artificial Intelligence
 - b) ML
 - c) Network Security
 - d) IT
10. The process of verifying the identity of a user.
- a) Authentication
 - b) Identification
 - c) Validation
 - d) Verification
11. An algorithm in encryption is called _____

- a) Algorithm
 - b) Procedure
 - c) Cipher
 - d) Module
12. The information that gets transformed in encryption is _____
- a) Plain text
 - b) Parallel text
 - c) Encrypted text
 - d) Decrypted text

Answers

1(b)	2(a)	3(d)	4(b)
5(b)	6(a)	7(d)	8(a)
9(c)	10(a)	11(c)	12(a)

Further Readings



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